IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MICROELECTROMECHANICAL SYSTEMS (MEMS)

Thursday, 15 May 2025, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer Question ONE, and any TWO of the remaining questions (Questions 2, 3, and 4).

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : Z. Durrani

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Information for Candidates

The following physical constants may be used:

Permittivity of free space: $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m

The change δL in span due to the arc length constraint for an axially unconstrained flexure of initial length L is given by:

$$\delta L \approx -\frac{3u^2}{5L}$$

where u is the lateral deflection.

The fractional resistance change of a <110>-aligned piezoresistor subject to longitudinal and transverse stresses σ_L and σ_T may be written as:

$$\frac{\Delta R}{R} = \sigma_L \pi_L + \sigma_T \pi_T$$

where $\pi_L = \frac{1}{2}\{\pi_{11} + \pi_{12} + \pi_{44}\}$ and $\pi_T = \frac{1}{2}\{\pi_{11} + \pi_{12} - \pi_{44}\}$, and π_{ij} are the standard piezoresistive coefficients.

The maximum stress σ_{max} in a square membrane of side L and thickness h subject to a uniformly distributed load p is given by:

$$\sigma_{max} \approx 0.3\left(\frac{L}{h}\right)^2 p$$

The quality factor of an in-plane micromechanical resonator damped by viscous losses in the gas between the moving element and the substrate is given by:

$$Q = \frac{m\omega_0 d}{\mu A}$$

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This question is compulsory.

1. a) Write down the bending equation for a buckled beam that has pinned joints at both ends. By solving the equation subject to appropriate boundary conditions, derive an expression for the critical axial load at which buckling will occur. [5]

b) Assuming damping is dominated by Couette flow between the proof mass and the substrate, derive scaling laws for the resonant frequency and Q (quality factor) of an in-plane micromechanical resonator in a scenario where the gap between mass and substrate is held constant while all other dimensions are scaled. [6]

c) A square silicon nitride membrane with a side length of 1.0 mm is to be fabricated by anisotropic etching a (100) silicon wafer that has been coated both sides with silicon nitride. The etchant is known to have an anisotropy of 35±5 and a <100> etch rate of 1.2±0.1 μm per minute.

i) If the wafer thickness is 380 μm, what size of mask aperture should be used in order to achieve the desired membrane size with the nominal etchant parameters? [4]

ii) If the etch process is run for long enough to guarantee the endpoint is reached, what are the maximum and minimum membrane sizes that might be obtained? [2]

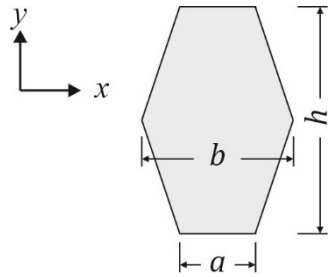
d) A bulk micromachined pressure sensor has a square silicon membrane of side length 2.0 mm and thickness 10 μm. The membrane stress is monitored using a single p-type piezo-resistor positioned at the mid-point of one of the membrane edges and oriented so as to maximise its response. What will be the fractional change in resistance caused by a pressure difference of 100 Pa across the membrane?

Assume the following values for the piezoresistive coefficients of silicon (all in units of 10^{-11} Pa^{-1}): π_{11} = 6.6, π_{12} = -1.1, π_{44} = 138.1. Poisson's ratio for the relevant directions is ν = 0.064. [6]

Question 1 continues on the next page.

e) A particular fabrication process with etching from both sides of the wafer produces beams with an irregular hexagonal cross-section as shown in Figure 1.1. Show that the second moment of area for a beam of this shape undergoing bending in the *xz*-plane is as shown on the right.

[5]



$$I = \frac{(a^2 + b^2)(a + b)h}{48}$$

Figure 1.1

f) Explain briefly the operation of a typical electrothermal shape bimorph actuator, illustrating your answer with sketches showing both undeflected and deflected shapes. Also derive an approximate expression for the tip deflection in terms of appropriate dimensions and the average temperature rises in the different sections.

[6]

g) Briefly compare and contrast electron beam lithography and focused ion beam as direct-write techniques, including their operation principles, resolution capabilities and applications.

[6]

End of Question 1

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Answer ANY TWO of the remaining questions (Questions 2, 3 and 4).

2. a) In the absence of damping, the lateral deflection $v(x, t)$ of a vibrating beam satisfies the wave equation:

$$EI\frac{\partial^4 v}{\partial x^4} + m\frac{\partial^2 v}{\partial t^2} = 0$$

where m is the mass per unit length of the beam, and E and I are respectively its Young's modulus and second moment of area.

Verify that the normal vibrational modes of such a beam are of the form:

$$v(x,t) = [A\cos(kx) + B\sin(kx) + C\cosh(kx) + D\sinh(kx)] \cdot \exp(j\omega t)$$

Also derive the mathematical relationship between ω and k. [8]

b) State the boundary conditions that will apply to the function $v(x, t)$ in the case of a beam that is built in at both ends. By applying these boundary conditions show that for such a beam k must satisfy the eigenvalue equation:

$$\cos(kL)\cosh(kL) = 1$$

where L is the length of the beam. [8]

c) A chemical sensor is fabricated from a built-in silicon beam that has a selectively absorbing, 0.5 μm-thick polymer coating covering its upper surface. The polymer has a density of 1500 kg/m^3 and its Young's modulus is negligible in comparison to that of silicon. If the cantilever is 10 μm wide, 500 μm long, and 3 μm deep, calculate the frequency of the sensor's lowest order vibrational mode. Also determine the frequency shift that will occur when the polymer absorbs 1 pico-gram of analyte. [8]

d) Show that, if the beam in part c) is vibrating in its lowest mode, and the vibration amplitude (zero to peak) at the centre is v_0, then the maximum axial strain near the anchor points will be given approximately by:

$$\varepsilon_{max} \sim 14\frac{hv_0}{L^2}$$

where h is the beam depth. (*hint:* you need to evaluate the curvature at the root of the beam at the peak of the oscillation cycle; from this you can obtain the maximum strain.) [6]

Assume values of 170 GPa and 2330 kg/m^3 for the Young's modulus and density of silicon. The smallest positive root of the equation $\cos(x)\cosh(x) = 1$ is $x = 4.73$.

3. a) Figure 3.1(a) shows a section of an electrostatic comb drive in which the moving electrodes can undergo displacement in both the *x*- and *y*- directions. Using the principle of virtual work, show that the *x*- and *y*- components of the actuator force experienced by the moving electrode array are:

$$F_x = N\varepsilon_0 hV^2\left(l+u_y\right)\frac{gu_x}{\left(g^2-u_x^2\right)^2} \quad ; \quad F_y = \frac{N\varepsilon_0 hV^2}{2}\frac{g}{g^2-u_x^2}$$

where l and g are the overlap length and inter-electrode gap when the applied voltage $V = 0$, $\boldsymbol{u} = \left(u_x, u_y\right)$ is the displacement of the moving electrode array, N is the number of electrode gaps, and h is the mechanical layer thickness. [10]

b) Figure 3.1(b) shows a MEMS device comprising a comb drive and a portal frame suspension. It is intended for use as a long-travel *y*-axis actuator.

By considering the mechanical equilibrium of each flexure, derive relations between R_{1x}, R_{1y} and C_1 and between R_{2x}, R_{2y} and C_2. Also, by considering the mechanical equilibrium of the entire structure, show that:

$$R_{1x} = R_{2x} = -\frac{1}{2}F_x$$

You may assume that, if $(0,0)$ is the geometric centre of the portal frame, then the line of action of $\left(F_x, F_y\right)$ always passes through the point $\left(0, 0.5u_y\right)$. [8]

c) Assuming u_x arises purely from the lateral shift of the moving electrodes due to the arc length constraint on the flexures, and neglecting quadratic and higher order terms in u_x, show that F_x may be expressed in the approximate form:

$$F_x \approx -\frac{6\left(l+u_y\right)u_y^2}{5g^2L}F_y$$

[8]

d) An actuator of the type shown in Figure 3.1(b) is built and tested, and it is found that the device is prone to mechanical instability at larger displacements. Considering the equations in parts a) to c), why do you think this behaviour arises? Can you suggest a simple modification to the design that would eliminate it? [4]

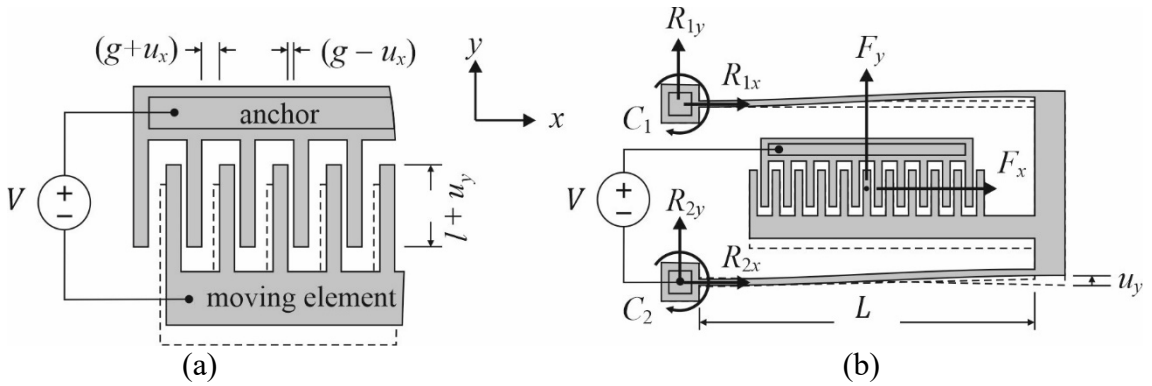


Figure 3.1

4. a) The transverse stiffness, k, of a flexure varies roughly linearly with axial load F (tensile if $F > 0$) decreasing to zero when the load reaches $-F_0$ where $F_0 = \frac{\pi^2 EI}{L^2}$. Here L is the flexure length and the other parameters have their usual meanings. Using this information, show that this dependence can be expressed approximately as:

$$k \approx \frac{12F_0}{\pi^2 L}\left(1 + \frac{F}{F_0}\right) \quad ; \quad F \geq -F_0$$

[4]

b) Show that the dependence discussed in part a) leads to a cubic non-linearity in certain types of MEMS suspension and explain briefly why this can be problematic. [6]

The remainder of this question concerns the device shown in Figure 4.1 which is a micromachined z-axis accelerometer fabricated in a nickel mechanical layer on a silicon substrate. The proof mass is suspended on four flexures.

c) One of the design objectives is to try and use thermally induced stress in the suspension to compensate for the variation with temperature of the Young's modulus of nickel. Explain why the device design results in the nickel stress rising with increasing temperature, and show that the change $\Delta\varepsilon_{th}$ in thermally induced strain due to a temperature rise ΔT is given by:

$$\Delta\varepsilon_{th} = \frac{L_s}{2L}(\alpha - \alpha_s)\Delta T$$

where α and α_s are the thermal expansion coefficients of nickel and silicon respectively. [8]

d) Using the results in parts a) and c), show that the desired temperature compensation will be achieved if the spacing L_s between the flexures is set according to:

$$\frac{L_s}{L} = -\frac{\pi^2}{6}\frac{h^2}{L^2(\alpha - \alpha_s)}\frac{1}{E}\frac{dE}{dT}$$

where h is the mechanical layer thickness, and the term $\frac{1}{E}\frac{dE}{dT}$ describes the temperature dependence of Young's modulus. [8]

e) If the flexures are 200 μm long, and the mechanical layer is 5 μm thick, calculate the required value for L_s. You may assume $\alpha = 13.3 \times 10^{-6}\ \mathrm{K}^{-1}$, $\alpha_s = 2.5 \times 10^{-6}\ \mathrm{K}^{-1}$, and $\frac{1}{E}\frac{dE}{dT} = -1.2 \times 10^{-3}\ \mathrm{K}^{-1}$. [4]

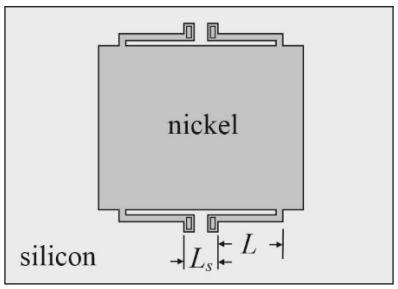


Figure 4.1